STAT 578: Topics in High Dimensional Statistics

Syllabus, Spring 2025

This version: January 19, 2025

High-dimensional statistical models usually concern the case where the number of features/parameters is large compared to the number of independent samples. Examples include Lasso, graphical Lasso, matrix estimation, and principal component analysis. There is some overlap in topics with STAT 542 Statistical Learning, but we hope to offer some different perspectives and mathematical tools for analysis of high-dimensional models that are often seen in research, such as approximate message passing, leave-one-out analysis, and convex geometry.

Lecture time and location:

01:00PM - 01:50PM MWF 203 Transportation Building

1 Contact Information

Instructor:

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2 References

The lecture notes from a previous iteration of the course can be found in `https://jingbol.web.illinois.edu/notes.pdf` There is no textbook for this course, but here are a few general references on the related topics:

Philippe Rigollet and Jan-Christian Hütter (2019). High Dimensional Statistics. (lecture notes available online; reference for Gaussian sequence model, linear regression, and matrix estimation)

Ramon van Handel. Probability in High Dimension `https://web.math.princeton.edu/~rvan/APC550.pdf` *(a consistent semigroup approach for concentration, empirical processes, and universality)*

Roman Vershynin. High-Dimensional Probability: An Introduction with Applications in Data Science `https://www.math.uci.edu/~rvershyn/papers/HDP-book/HDP-book.pdf` *(intro to concentration of sub-Gaussian and sub-exponential distributions; random matrix theory matrix)*

Martin J. Wainwright (2019). High-Dimensional Statistics: A Non-Asymptotic Viewpoint.

Simon Foucart and Holger Rauhu (2013). A Mathematical Introduction to Compressive Sensing. (Springer)

Larry Wasserman (2005). All of Nonparametric Statistics.

3 Grading

- Homework (50%) We plan to leave 3 hw with the following due dates
  Feb 28 [10%] exercises on concentration of sub-Gaussian and sub-exponential variables, following Vershynin
  Apr 4 [20%]
  Apr 25 [20%]

  For hw2 and hw3, the exercises can be found in the end of my lecture notes `https://jingbol.web.illinois.edu/notes.pdf` You are not required to learn all the topics, and hence not required to solve all the problems. Number of credits are given next to each problem. For hw2 and hw3, if you submit solutions to exercises with total points $a_1, a_2, \dots$, we will grade the first n problems where $n = \min\{i\colon a_1 + \cdots + a_i \geq 20\}$, getting the scores $b_1, \dots, b_n$, and your grade for that hw will be $\min\{b_1 + \cdots + b_n, 20\}$.

- Midterm (25%) Take home exam during the midterm week (from Mon to Fri).

- Final project (25%)
  Form groups of size 3-5

  Each group focus on one topic in Section 7, or email me to propose alternative topics.

  Ideally each person focus on at least one paper under the topic, and have weekly discussions with other team members about other papers in the same topic.

  Focus on: why is the subject important; what are the existing techniques and results; **what are the major open problems in this direction.**

  Write a 10-page report. If you write more than 10 pages, we will only read the first 10 pages for grading.

  Group presentation during the last week of the semester.

4 Topics Covered

We hope to cover a number of popular models or algorithms in high-dimensional statistical inference, such as

- Lasso
- graphical Lasso
- matrix estimation
- principal component analysis

Moreover, a few techniques for analysis of these models will be introduced. These may include (depending on time and interests)

- information-theoretic (minimax) lower bounds
- restricted eigenvalue conditions
- nullspace conditions, statistical dimension and other convex geometric methods
- statistical physics predication (replica analysis)
- iterative algorithms, such as iterative soft/hard thresholding
- approximate message passing.

5 Weekly Schedule

1. Introduction; concentration of random variables
2. Gaussian sequence model
3. Lasso; restricted eigenvalue condition; fast rate
4. Nullspace condition
5. Statistical dimensions and convex geometric tools
6. Graphical Lasso
7. Matrix estimation
8. Robust PCA
9. Information-theoretic technique for lower bounds
10. Empirical distribution of error: leave-one-out
11. Empirical distribution of error: replica analysis

12. Iterative soft/hard thresholding, approximate message passing

2-5 are mostly sparse linear regression or the related. 3-4 introduce some tools for analyzing Lasso and provide guarantees on the risk. 9 is about general ideas of showing lower bounds on the risk. 10, 11 are techniques for more refined analysis that even provides the empirical distribution of the errors, and 12 additional algorithms for sparse regression. 10, 11, 12 are conceptually more challenging than the others in the list, but are also of greater interest in recent research (in my opinion). Papers on these topics are generally challenging to read, which, in my opinion, is partly because those authors' primary goals are to present original results to get credits. In this course, however, my goal is tutorial, so I will "deconstruct" their technique by working on the simplest model possible. Nevertheless, you are not required to understand deeply all the topics in the limited time (see the grading instructions).

6 Required background

Basic courses in probability and statistics are assumed. You should also know basic notions of linear algebra, such as eigenvalue decomposition or singular value decomposition.

7 Readings Project Examples Topics

1. Topic: geometry and high-dimensional statistics

   - D. Amelunxen, M. Lotz, M. McCoy, J. Tropp, Living on the edge: Phase transitions in convex programs with random data, Information and Inference, 2014.
   - D. Donoho and J. Tanner, Neighborliness of randomly projected simplices in high dimensions, PNAS, 2005.
   - T. Cover, Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition, IEEE trans. on electronic computers, 1965.
   - Chen, Y. (2021). An almost constant lower bound of the isoperimetric coefficient in the KLS conjecture. Geometric and Functional Analysis, 31, 34-61.
   - Chen, Yuansi, and Ronen Eldan. "Localization schemes: A framework for proving mixing bounds for Markov chains." 2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS). IEEE, 2022.

2. Topic: high-dimensional regression and matrix estimation

   - P. Sur, Y. Chen, and E. Candes, The likelihood ratio test in high-dimensional logistic regression is asymptotically a rescaled chi-square, 2017.
   - I. Daubechies M. Defrise C. De Mol, An iterative thresholding algorithm for linear inverse problems with a sparsity constraint `https://onlinelibrary.wiley.com/doi/abs/10.1002/cpa.20042`

- Benjamin Recht, Weiyu Xu, Babak Hassibi, Null Space Conditions and Thresholds for Rank Minimization https://people.eecs.berkeley.edu/~brecht/papers/10.RecXuHas.Thresholds.pdf
- Noureddine El Karoui, Derek Bean, Peter Bickel, Chingway Lim and Bin Yu, On robust regression with high-dimensional predictors https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.294.5920&rep=rep1&type=pdf
- Noureddine El Karoui,, On the impact of predictor geometry on the performance on high-dimensional ridge-regularized generalized robust regression estimators, Probab. Theory Relat. Fields (2018) 170:95-175

3. Topic: High-dimensional statistics and physics techniques

- Mohsen Bayati, Andrea Montanari, The dynamics of message passing on dense graphs, with applications to compressed sensing https://arxiv.org/pdf/1001.3448.pdf
- Yash Deshpande, Andrea Montanari, Information-theoretically Optimal Sparse PCA https://arxiv.org/pdf/1402.2238.pdf
- The replica trick for the analysis of random matrices https://meisong541.github.io/jekyll/update/2019/08/04/Replica_method_1.html#ref1
- Fan, Z., Mei, S., & Montanari, A. (2021). TAP free energy, spin glasses and variational inference.
- Zhong, Xinyi, Chang Su, and Zhou Fan. "Empirical Bayes PCA in high dimensions." Journal of the Royal Statistical Society Series B: Statistical Methodology 84.3 (2022): 853-878.
- Huang, Brice, and Mark Sellke. "A Constructive Proof of the Spherical Parisi Formula." arXiv preprint arXiv:2311.15495 (2023).
- Brice Huang, "Capacity threshold for the Ising perceptron" https://www.bricehuang.com/files/slides/ising-perceptron.pdf
- Ghio, D., Dandi, Y., Krzakala, F., & Zdeborova, L. (2024). Sampling with flows, diffusion, and autoregressive neural networks from a spin-glass perspective. Proceedings of the National Academy of Sciences, 121(27), e2311810121.

4. Topic: information-theoretic lower bounds

- Narayana P. Santhanam, Martin J. Wainwright, Information-Theoretic Limits of Selecting Binary Graphical Models in High Dimensions
- lecture notes on methods for statistical lower bounds via information theoretic techniques http://web.stanford.edu/class/ee378c/index.html
- Yuchen Zhang, Distributed machine learning with communication constraints https://www2.eecs.berkeley.edu/Pubs/TechRpts/2016/EECS-2016-47.pdf
- Khribch, EL Mahdi, and Pierre Alquier. "Convergence of Statistical Estimators via Mutual Information Bounds." arXiv preprint arXiv:2412.18539 (2024).

Feel free to discuss any other papers/monographs you are interested in and would like to present.